The following problem involves the number a_3, which is given by a computer code that represents a_3 as a computable number.

The code for a_3 is:

```
"Compute_a_3(n)
Input : natural number n
Output = 1 + (1/2)^n - 72^(-n)
end
"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_3 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^{∞} norm, if a_3 is given as in the codes above. Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 190 seconds >

The code defines a_3 as the limit of the sequence $1 + 2^{-n} - 72^{-n}$ as $n \to \infty$. Since $2^{-n} \to 0$ and $72^{-n} \to 0$, the limit is exactly $a_3 = 1$.

The problem is to minimize total treatment time $x_1 + x_2$ subject to $a_1 x_1 + a_3 x_2 = 1$ with $a_1 = 1$, $a_3 = 1$, and $x_1, x_2 \geq 0$. This reduces to minimizing $x_1 + x_2$ subject to $x_1 + x_2 = 1$. The objective is constantly 1, so every pair (x_1, x_2) with $x_1 + x_2 = 1$ and $x_1, x_2 \geq 0$ is an exact optimal minimizer.

To give an approximation to at least 2 digits accuracy in the ℓ^{∞} norm, we can simply provide an exact minimizer:

$$x_1 = 0.50, \quad x_2 = 0.50$$

(Any other nonnegative pair summing to 1, such as $x_1 = 0.00, x_2 = 1.00$, is equally valid.)